

EXAM IN: STA510 STATISTICAL MODELING AND SIMULATION

DURATION: 4 HOURS

DATE: February 23th, 2026

PERMITTED AIDS: Approved simple calculator

One A4 size sheet with your own handwritten notes.

THE EXAM CONSISTS OF 4 PROBLEMS ON 3 PAGES, 6 PAGES OF ENCLOSURES.

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Problem 1

Let X be a real-valued random variable with density function function

$$f(x) = \frac{1 + \sin(x)}{\pi + 2}, \quad 0 \leq x \leq \pi,$$

and $f(x) = 0$ otherwise.

- a) Show that $f(x)$ is a proper probability density function. (4P)
- b) Find the cumulative density function $F(x)$, and verify that inverse transform method is not suitable to generate i.i.d. samples from $f(x)$. (5P)
- c) We wish to generate i.i.d. samples from $f(x)$ using the Accept-Reject (AR) method with a Uniform proposal.
 - i) Specify the proposal distribution, and find the envelope constant c for the AR method. (5P)
 - ii) Write down a the AR algorithm (in pseudocode) to generate one draw from $f(x)$ using your proposal distribution and constant c . (4P)

Problem 2

Consider the following R code:

```
T <- 1000
x <- numeric(T)
x[1] <- 0

for (t in 2:T) {
  x[t] <- x[t - 1] + sample(c(-1, 1), size = 1, prob = c(0.5, 0.5))
}
```

- a) i) Identify what kind of stochastic process this R code simulates. Justify your answer. (3P)
ii) Is this a Markov process? Justify your answer! (3P)
iii) How would the process change if you modify $c(0.5, 0.5)$ to $c(0.4, 0.6)$? (3P)
- b) Write down an expression for the stochastic process shown in the above R code. What is the state space of this process?
Discuss whether this process has a stationary distribution. Justify your reasoning analytically by finding the expectation and variance of this process given $E(X_1) = 0$. (12P)

Problem 3

We want to estimate the mathematical constant π based on the property that $P(X^2 + Y^2 \leq 1) = \frac{\pi}{4}$ for $X \sim U[0, 1]$ and $Y \sim U[0, 1]$. For this we have implemented two functions in the following R-code:

```
# Option 1:
estPi1 <- function(n=10000){
  y <- numeric(n)
  for(i in 1:n){
    u <- runif(2)
    y[i] <- (u[1]^2 + u[2]^2 <= 1)
  }
  return(4*mean(y))
}

# Option 2:
estPi2 <- function(n=10000){
  u <- matrix(runif(2*n), nrow=n, ncol=2, byrow = T)
  y <- mean(rowSums(u^2) <= 1)
  return(4*y)
}
```

- a) Which algorithm has been implemented for `estPi1()`? Reason your answer, and describe the choices made for the algorithm.
Write down an expression for the estimator $\hat{\pi}$ of π (8P).
- b) Do both of the two above R-functions `estPi1()` and `estPi2()`, return a correct estimate for π ? Justify your answer. (3P)
- c) Explain how we can improve `estPi1()` to estimate π , and describe the necessary changes in the R-code. (5P)

Problem 4

Assume we are monitoring the air pollution in Stavanger day by day, and each day we collect multiple sensor readings $y_1, \dots, y_n$ where y_i describes the concentration of the air pollution at time point $i = 1, \dots, n$, and n is the total number of readings for today.

Now, we want to estimate for the current day, the true (unobserved) mean pollution level μ with within-day variability σ^2 while incorporating prior knowledge (from past days) about what typical pollution levels and variances look like. It is assumed that

$$y_i \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2), \quad i = 1, \dots, n.$$

For the prior information about μ and σ^2 , we are assuming that the mean population level

$$\mu \mid \sigma^2 \sim N(m_0, \sigma^2/\kappa_0)$$

with known $m_0, \kappa_0 > 0$, and for the variance we are assuming

$$\log(\sigma^2) \sim N(0, 1)$$

independently of μ .

- a) Show that the joint posterior kernel $p(\mu, \sigma^2 \mid y_1, \dots, y_n)$ is given by

$$p(\mu, \sigma^2 \mid y_1, \dots, y_n) \propto \frac{1}{(\sigma^2)^{\frac{(n+3)}{2}}} \exp \left(-\frac{1}{2\sigma^2} \left(S(\mu) + \kappa_0(\mu - m_0)^2 \right) - \frac{(\log(\sigma^2))^2}{2} \right)$$

with $S(\mu) = \sum_{i=1}^n (y_i - \mu)^2$. (8P)

- b) Derive the conditional log posterior kernel for $\log(p(\mu \mid \sigma^2, y_1, \dots, y_n))$, and show that the conditional posterior $\mu \mid \sigma^2, y_1, \dots, y_n \sim N(m_n, \sigma^2/\kappa_n)$. Find m_n and κ_n ! (8P)
- c) Derive the conditional log posterior kernel for $\log(p(\sigma^2 \mid \mu, y_1, \dots, y_n))$, and explain why it is not a standard distribution we can sample from directly. (4P)
- d) Let us consider a Metropolis–Hastings step for the posterior kernel for $\sigma^2 \mid \mu, y_1, \dots, y_n$ with proposal

$$\log(\sigma^2) \sim N(\sigma_{t-1}^2, s^2).$$

Specify the proposal distribution $q(\cdot \mid \cdot)$, and derive the Metropolis–Hastings acceptance probability $\log(\alpha)$. (6P)

- e) Describe briefly the Gibbs sampler algorithm to simulate values for $p(\mu, \sigma^2 \mid y_1, \dots, y_n)$. (4P)